

Embedded Linux Kernel Optimization

Boot Time Reduction and Performance Tuning

Project README

22.287 s → 7.527 s

Userspace startup time before and after optimization (about 66% reduction)

1. About the Project

Embedded Linux systems such as routers, IoT devices, automotive units and industrial controllers need to start quickly. This project studies how a Linux system boots, measures the boot time with the built-in `systemd-analyze` tools, identifies slow startup units, and applies a shell script that disables non-essential services (Bluetooth, CUPS printing and Avahi network discovery). The boot time is measured again after a reboot and the results are compared.

2. Repository Contents

File / Folder	Description
README.pdf / README.md	Project overview (this document)
docs/Project_Report.pdf	Full report: theory, method, screenshots, results and conclusion
docs/Scripts_and_Commands.pdf	All commands and scripts with line-by-line explanation
scripts/optimize_services.sh	Disables bluetooth, cups and avahi-daemon
scripts/boot_analysis.sh	Prints boot time, slowest services, critical chain, memory and kernel log
screenshots/	Terminal screenshots taken before and after optimization, plus comparison chart

3. Requirements

- Ubuntu Linux (native, virtual machine, or WSL with `systemd` enabled)
- `systemd` (provides `systemd-analyze` and `systemctl`), Bash, and `sudo` access
- Optional for full embedded work: Linux kernel source, build-essential, and a development board with a serial console

4. How to Run

```
git clone https://github.com/praveenmanne09/Embedded-Linux-Kernel-Optimization.git
cd Embedded-Linux-Kernel-Optimization

chmod +x scripts/*.sh
systemd-analyze                # 1. baseline boot time
systemd-analyze blame | head -10 # 2. slowest services
./scripts/optimize_services.sh # 3. disable unneeded services
sudo reboot                    # 4. reboot
systemd-analyze                # 5. measure again and compare
```

Only disable services your system does not need. For example, do not disable networking services if your project depends on the network. A disabled service can be restored with `sudo systemctl enable <service>`.

5. Results Summary

Metric	Before	After
Startup finished (userspace)	22.287 s	7.527 s
graphical.target reached	21.861 s	7.381 s
Slowest unit	dev-sdd.device (4.242 s)	landscape-client.service (3.007 s)

See **docs/Project_Report.pdf** for screenshots, the full service-by-service comparison and notes on how to interpret the numbers.

6. Author

Praveen Manne

GitHub: github.com/praveenmanne09

LinkedIn: [linkedin.com/in/praveen-manne-1b8902269](https://www.linkedin.com/in/praveen-manne-1b8902269)